PRODUCT TEARDOWN

Sentinel Cyber UW Co-Pilot

How AI augments underwriting judgment without replacing it

WHAT'S INSIDE

→ Product architecture & core capabilities

→ AI eval framework (4-tier rubric)

→ Design principles: transparency + trust

→ What's working & what's missing

68%
Faster triage

12x
More flags

< 3min
AI decision

THE PROBLEM

Cyber underwriting is broken by fragmentation

Document Overload

Underwriters drown in SOC reports, questionnaires, audit PDFs — with no structured ingestion layer

Fragmented Signals

Risk signals live across emails, scans, external feeds. No single source of truth

Speed vs Precision

Manual stitching = slow decisions. Rushing = adverse selection risk

Portfolio Blindness

No real-time view of appetite alignment or accumulation risk

Reality: Speed + Precision expected. Fragmented workflows that don't scale — delivered.

ARCHITECTURE

5-Layer Intelligence Stack

01
Submission Ingestion
OCR + LLM

Emails, PDFs, scans → structured data. Completeness check from day one.

02
Cyber Risk Intelligence
RAG Engine

Vuln signals, incident history, cloud exposure, 3rd-party deps correlated.

03
Guideline Alignment
ML Model

Validates against insurer rules, exclusions, thresholds in real-time.

04
Smart Triage Engine
ML Scoring

Prioritises high-quality submissions. Risk-fit score before underwriter rev...

05
Decision Summaries
LLM Synthesis

Single underwriting view. Eliminates manual stitching across sources.

STANDOUT FEATURE

Red Flags Engine

The AI doesn't just read — it cross-examines

LIVE EXAMPLE: Document Conflict Detection

PROPOSAL FORM SAYS

"Daily offline backups, verified weekly"

Section 4.2 · Backup Policy

VS

AUDIT REPORT SAYS

"No restore test since 2021 — EXPIRED"

Pg 14 · Audit Report

SENTINEL FLAGS: **Backup Verification Conflict → HIGH RISK**

Evidence chain: Proposal Form (S4.2) contradicts Audit Report (Pg 14). Confidence: 97%

Core Cyber
5 Flags

Document
3 Flags

Submission
2 Flags

AI EVALS FRAMEWORK

The 4-Tier Rubric

T1
TIER

Output Quality

- Precision@K on flagged risks
- Hallucination rate on doc cross-refs

T2
TIER

Reasoning & Explanation

- Evidence chain completeness
- Source attribution accuracy

T3
TIER

Human-AI Collaboration

- Override rate + confidence delta
- Time-to-decision improvement

T4
TIER

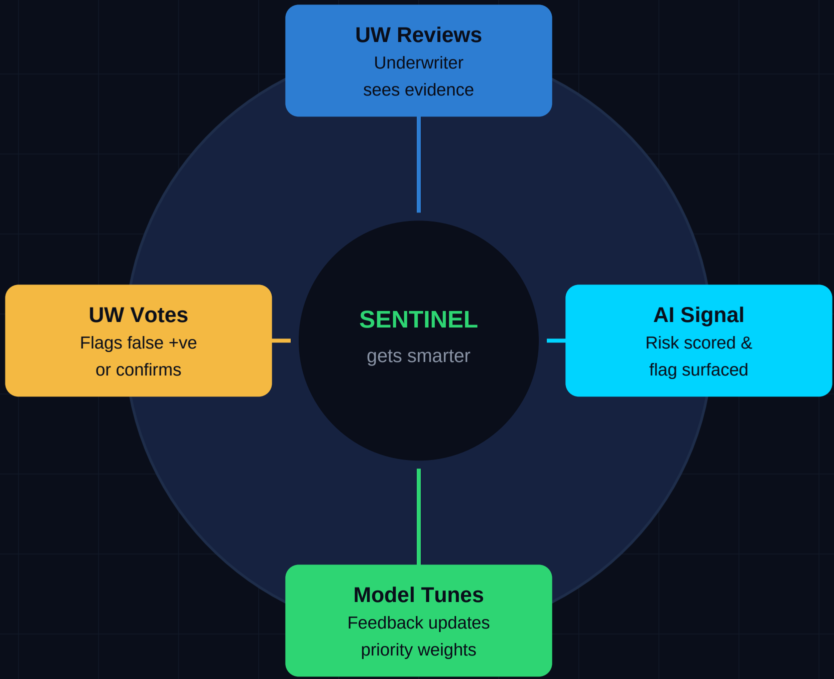
Portfolio Outcomes

- Loss ratio: AI-assisted vs manual
- Adverse selection reduction

TRUST MECHANISM

The Feedback Loop

The most underrated eval mechanism in production AI



Key Insight: Underwriter judgment becomes the training signal. Trust compounds over time.

DESIGN PRINCIPLES

The AI Must Show Its Work

Evidence First

Every flag cites its source document, page, and section. No black-box conclusions.

Calibrated Confidence

Show confidence %. An underwriter handles a 60% flag differently than a 97% one. That delta matters.

Override by Design

AI recommends. Underwriter decides. Override is not a failure state — it's the mechanism.

Uncertainty as Signal

Where the AI is unsure, it says so. Silence ≠ certainty. Flagging gaps is a feature.

Audit Trail Always

Every AI-assisted decision is logged with the model version, confidence, and human action taken.

OPEN QUESTIONS

What's Still Hard

Building the co-pilot is 50%. Here's the other half.

? How do you set confidence thresholds for 'good enough to act on'?

? How do you prevent underwriters from over-trusting scores over time?

? How do you benchmark a model that keeps evolving with feedback?

? How do you measure completeness vs hallucination risk at portfolio scale?

Explore Sentinel → uwcopilot-cyberinsurance.vercel.app

Drop your thoughts below · Follow for Part 3: Operationalizing AI in UW teams